

Brandt’s conjecture on regularizable dense triangle-free graphs follows from the Strong Brandt–Thomassé theorem

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July 23, 2026

Abstract

Brandt conjectured in 2002 that every maximal triangle-free graph G whose fractional total domination number satisfies $d_f(G) < 3$ admits a nonnegative rational solution of $A(G)x = \mathbf{1}$. We make two observations. First, the conjecture follows from ingredients already present in the literature: the Brandt–Thomassé structure theorem for maximal triangle-free graphs of minimum degree exceeding $n/3$, combined with the explicit equality weights on Andrásfai and Vega graphs going back to Brandt–Pisanski and Brandt–Thomassé (recorded in the published 2021 work of Łuczak, Polcyn, and Reiher), via a short rational blow-up argument; the same deduction settles Conjectures 4.1, 5.2, and 5.3 of Brandt’s paper. These resolutions appear not to have been stated explicitly before. Second, conditional on Theorem 1.2 of the 2024 preprint *Strong Brandt–Thomassé Theorems* (arXiv:2406.10745), a linear-programming duality argument upgrades this to an exact characterization: for maximal triangle-free graphs, $d_f(G) < 3$, the first-order property D_4 , and membership in the class of Andrásfai/Vega blow-ups all coincide, and the equality solution may always be taken strictly positive. The strict threshold is exact: a 9-vertex maximal triangle-free graph with $d_f = 3$ admits nonnegative but no positive equality solutions, refuting the non-strict variant recorded on D. West’s open problems page. All external structural inputs remain unrefereed (an unpublished manuscript and a 2024 preprint); no priority claim is made, and specialist review is requested.

1 Introduction

All graphs in this note are finite and simple. For a graph G with adjacency matrix $A = A(G)$ and all-ones vector $\mathbf{1}$, the *fractional total domination number* is

$$d_f(G) = \min\{\mathbf{1}^T p : p \in \mathbb{R}_{\geq 0}^{V(G)}, Ap \geq \mathbf{1}\}, \quad (1)$$

with LP dual

$$d_f(G) = \max\{\mathbf{1}^T z : z \in \mathbb{R}_{\geq 0}^{V(G)}, A^T z \leq \mathbf{1}\}. \quad (2)$$

Since $A = A^T$, the transpose is retained only to display the standard pairing. For a graph with no isolated vertex both programs are feasible and bounded, strong duality holds, and rational optimal certificates exist. For the one-vertex graph the primal program is infeasible and $d_f = \infty$ by convention; maximal triangle-free graphs on at least two vertices have no isolated vertices, so all statements below are nonvacuous exactly there.

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A graph is *maximal triangle-free* (MTF) if it is triangle-free and the addition of any missing edge creates a triangle; equivalently, every two nonadjacent vertices have a common neighbour.

In [1, Conjecture 5.1], Brandt made the following conjecture; the statement below is verbatim up to typography, and his equation (3) defines d_f exactly as in (1).

Conjecture 1.1 (Brandt [1, Conjecture 5.1]). *If G is a maximal triangle-free graph with $d_f(G) < 3$ then $A(G)x = \mathbf{1}$ has a rational solution $x \geq 0$.*

Brandt introduced Conjecture 1.1 as a linear-programming relaxation of his regular-supergraph conjecture [1, Conjecture 4.1]: every maximal triangle-free graph with minimum degree *exceeding* $n/3$ has a regular supergraph obtained by vertex multiplications. He further observed [1, p. 43] that on a *core* (twin-free graph), every nonnegative rational solution of $Ax = \mathbf{1}$ under $d_f(G) < 3$ is automatically strictly positive; the proof below delivers positivity directly for the full hypothesis class.

The conjecture connects an optimization threshold to a regularization property: positive integer multiplicities m_v with $Am = d\mathbf{1}$ describe a vertex multiplication of G into a d -regular graph, and scaling shows such a multiplication exists exactly when $Ax = \mathbf{1}$ has a positive rational solution. A non-strict variant of the statement, with hypothesis $\delta(G) \geq |V(G)|/3$ and positive multiplicities, is recorded on West’s open problems page [3]; see Section 9.

The external input to our argument is the classification theorem of Łuczak, Polcyn, and Reiher [2]. All statements below that depend on it are conditional on the correctness of that preprint, which is unrefereed at the time of writing.

2 The classification input

We transcribe the two statements of [2] that we use.

Definition 2.1 ([2, Definition 1.1]). For $k \in \mathbb{N}$, a graph G has property D_k if for every $m \in [k] = \{1, \dots, k\}$ and every sequence $x_1, \dots, x_{3m} \in V(G)$ of not necessarily distinct vertices there exists $y \in V(G)$ with

$$|\{j \in [3m] : x_j y \in E(G)\}| \geq m + 1.$$

Adjacency is thus counted with sequence multiplicity, and D_4 quantifies over $m \in \{1, 2, 3, 4\}$.

For a base graph H , a *blow-up* of H replaces each vertex v by a nonempty independent set C_v , with complete bipartite graphs between C_u and C_v exactly when $uv \in E(H)$. Every class is nonempty; vertex deletions are not blow-ups.

Theorem 2.2 ([2, Theorem 1.2]). *A maximal triangle-free graph satisfies D_4 if and only if it is a blow-up of either an Andrásfai or a Vega graph.*

The Andrásfai and Vega graphs are recalled in Sections 4 and 5, following the constructions in [2].

3 Failure of D_4 gives a dual certificate of value three

Lemma 3.1. *Let G be a finite graph. If G fails D_4 , then $d_f(G) \geq 3$. Equivalently, $d_f(G) < 3$ implies that G satisfies D_4 .*

Proof. If G fails D_4 , there are $m \in \{1, 2, 3, 4\}$ and a sequence $x_1, \dots, x_{3m} \in V(G)$ such that every $y \in V(G)$ satisfies

$$|\{j \in [3m] : x_j y \in E(G)\}| \leq m. \quad (3)$$

Let $c_v = |\{j : x_j = v\}|$ be the multiplicity of v , and set $z_v = c_v/m$. Then $z \geq 0$ and $\mathbf{1}^T z = \frac{1}{m} \sum_v c_v = \frac{3m}{m} = 3$. For every y ,

$$(A^T z)_y = \sum_{v \in N(y)} \frac{c_v}{m} = \frac{1}{m} |\{j : x_j y \in E(G)\}| \leq 1$$

by (3). So z is feasible for (2) with value 3, and weak duality gives $d_f(G) \geq 3$. $\square$

Remark 3.2. The abstract of [2] highlights $m \in \{2, 3, 4\}$, while Definition 2.1 formally includes $m = 1$; Lemma 3.1 covers all four values, so no mismatch arises. For MTF graphs on at least two vertices D_1 is automatic anyway: in a sequence of three vertices, either some vertex repeats (any neighbour of it meets two positions) or the three are distinct, hence not a triangle, so two of them are nonadjacent and maximality provides a common neighbour.

4 Andrásfai graphs

For $i \geq 1$ the Andrásfai graph Γ_i has vertex set $\mathbb{Z}/(3i-1)\mathbb{Z}$, with $r \sim s$ exactly when $r - s \pmod{3i-1} \in \{i, \dots, 2i-1\}$. It is i -regular on $3i-1$ vertices and maximal triangle-free. Setting $x_v = 1/i$ for every v gives $Ax = \mathbf{1}$, $x > 0$; the same vector is dual feasible since $A^T x = Ax = \mathbf{1}$, so it is simultaneously primal and dual optimal and

$$d_f(\Gamma_i) = \frac{3i-1}{i} = 3 - \frac{1}{i} < 3. \quad (4)$$

5 Vega graphs and explicit equality weights

Fix $i \geq 2$. Following [2], start with an inner copy of Γ_i on $\{0, 1, \dots, 3i-2\}$ with the three independent colour classes

$$\mathcal{R} = \{0, \dots, i-1\}, \quad \mathcal{G} = \{i, \dots, 2i-1\}, \quad \mathcal{B} = \{2i, \dots, 3i-2\}.$$

Add an external hexagon $a - v - c - u - b - w - a$ and two further vertices x, y , with

$$N(a) \cap V(\Gamma_i) = N(u) \cap V(\Gamma_i) = \mathcal{R}, \quad N(b) \cap V(\Gamma_i) = N(v) \cap V(\Gamma_i) = \mathcal{G}, \quad N(c) \cap V(\Gamma_i) = N(w) \cap V(\Gamma_i) = \mathcal{B},$$

$$N(x) \cap \{a, b, c, u, v, w, y\} = \{a, b, c, y\}, \quad N(y) \cap \{a, b, c, u, v, w, x\} = \{u, v, w, x\}.$$

This graph is Υ_i^{00} ; the variants are $\Upsilon_i^{10} = \Upsilon_i^{00} - y$, $\Upsilon_i^{01} = \Upsilon_i^{00} - (2i-1)$, and $\Upsilon_i^{11} = \Upsilon_i^{00} - \{y, 2i-1\}$, so y is present iff $\mu = 0$ and the inner vertex $2i-1$ is present iff $\nu = 0$. Then $|V(\Upsilon_i^{\mu\nu})| = 3i + 7 - \mu - \nu$. Set

$$D = D_{\mu\nu} = 9i - 6 - \mu - \nu.$$

The weights below are not new: they originate in the Brandt–Thomassé manuscript [4], the underlying regular-blow-up observation goes back to Brandt–Pisanski [5], and both are recorded, with the same table, in [6, Section 4.1, Fact 4.2 and Figure 4.2]. We restate them and verify the identities in a self-contained way.

Theorem 5.1 (Brandt–Thomassé; see [6, Fact 4.2]). *For every $i \geq 2$ and $\mu, \nu \in \{0, 1\}$ there is a positive integer vector $w = w_i^{\mu\nu}$ on $V(\Upsilon_i^{\mu\nu})$ with*

$$Aw = D \mathbf{1}, \quad \mathbf{1}^T w = 3D - 1.$$

Consequently $q = w/D$ is a positive rational solution of $Aq = \mathbf{1}$, it is simultaneously primal and dual optimal, and

$$d_f(\Upsilon_i^{\mu\nu}) = 3 - \frac{1}{D} < 3.$$

Proof. Define w as follows. Inner vertices: if $\nu = 0$, put $w_0 = w_{2i-1} = 1$ and $w_j = 3$ for all other inner j ; if $\nu = 1$ (vertex $2i - 1$ deleted), put $w_0 = w_{i-1} = 2$ and $w_j = 3$ otherwise. External vertices, in the order (a, v, c, u, b, w, x, y) :

(μ, ν)	a	v	c	u	b	w	x	y
$(0, 0)$	$3i-2$	$3i-2$	$3i-3$	$3i-2$	$3i-2$	$3i-3$	1	1
$(1, 0)$	$3i-2$	$3i-3$	$3i-3$	$3i-3$	$3i-2$	$3i-4$	2	–
$(0, 1)$	$3i-2$	$3i-3$	$3i-3$	$3i-2$	$3i-3$	$3i-3$	1	1
$(1, 1)$	$3i-2$	$3i-4$	$3i-3$	$3i-3$	$3i-3$	$3i-4$	2	–

All entries are positive for $i \geq 2$ (the smallest is $3i - 4 \geq 2$).

Write $T_{\mathcal{C}}$ for the total inner weight of colour class $\mathcal{C}$ and $S_{\mathcal{C}}$ for the inner-neighbourhood weight of an inner vertex of class $\mathcal{C}$ (well defined by the circulant structure and the placement of the special weights). A direct computation gives, for $\nu = 0$:

$$(T_{\mathcal{R}}, T_{\mathcal{G}}, T_{\mathcal{B}}) = (3i-2, 3i-2, 3i-3), \quad (S_{\mathcal{R}}, S_{\mathcal{G}}, S_{\mathcal{B}}) = (3i-2, 3i-2, 3i),$$

using that each red vertex is adjacent to $2i - 1$ but not 0, each green vertex to 0 but not $2i - 1$, and each blue vertex to neither. For $\nu = 1$:

$$(T_{\mathcal{R}}, T_{\mathcal{G}}, T_{\mathcal{B}}) = (3i-2, 3i-3, 3i-3), \quad (S_{\mathcal{R}}, S_{\mathcal{G}}, S_{\mathcal{B}}) = (3i-3, 3i-1, 3i-1),$$

using that red vertices lose the neighbour $2i - 1$, while green (resp. blue) vertices are adjacent to 0 (resp. $i - 1$), of special weight 2.

The equation $(Aw)_z = D$ then verifies, class by class, as an affine identity in i . We display the variant $(\mu, \nu) = (0, 0)$, where $D = 9i - 6$: for inner vertices,

$$S_{\mathcal{R}} + w_a + w_u = S_{\mathcal{G}} + w_b + w_v = (3i-2) \cdot 3 = D, \quad S_{\mathcal{B}} + w_c + w_w = 3i + 2(3i-3) = D;$$

for the hexagon (each hexagon vertex sees its colour class total, its two hexagon neighbours, and one of x, y),

$$T_{\mathcal{R}} + w_v + w_w + w_x = T_{\mathcal{G}} + w_a + w_c + w_y = \dots = (3i-2) + (3i-2) + (3i-3) + 1 = D,$$

and for the outer vertices,

$$(Aw)_x = w_a + w_b + w_c + w_y = (Aw)_y = w_u + w_v + w_w + w_x = (3i-2) + (3i-2) + (3i-3) + 1 = D.$$

The totals are $\mathbf{1}^T w = (9i - 7) + (18i - 12) = 27i - 19 = 3D - 1$. The three remaining variants verify by the same routine computation with their tables; every neighbourhood equation and both totals are affine identities in i , and all four variants have been machine-checked for $i = 2, \dots, 8$ together with maximal triangle-freeness of the constructed graphs (script `verify/vega.weights.py` in the ancillary files).

Finally, $q = w/D > 0$ satisfies $Aq = \mathbf{1}$, hence is primal feasible for (1) with value $(3D - 1)/D$ and dual feasible for (2) with the same value, so $d_f = 3 - 1/D$. $\square$

6 Blow-ups and false twins

Lemma 6.1. *Let G be a blow-up of H with nonempty classes $(C_v)_{v \in V(H)}$. For p on $V(G)$ define $q_v = \sum_{a \in C_v} p_a$. Then $(A_G p)_a = (A_H q)_v$ for every $a \in C_v$, and $\mathbf{1}^T p = \mathbf{1}^T q$. Consequently:*

1. $d_f(G) = d_f(H)$;
2. $A_G p = \mathbf{1}$, $p \geq 0$ is solvable iff $A_H q = \mathbf{1}$, $q \geq 0$ is solvable;
3. $A_G p = \mathbf{1}$, $p > 0$ is solvable iff $A_H q = \mathbf{1}$, $q > 0$ is solvable.

Proof. The displayed identity is immediate from the blow-up structure. Feasible vectors on G give feasible class-sum vectors on H with the same objective; conversely distribute q_v uniformly over C_v , $p_a = q_v/|C_v|$, which preserves feasibility and objective by the identity. For (3), positivity of class sums descends trivially, and the uniform lift is strictly positive precisely because every $|C_v| \geq 1$. Rationality is preserved in both directions. $\square$

Lemma 6.2. *If H is maximal triangle-free with at least two vertices, then every blow-up of H is maximal triangle-free. Conversely, every maximal triangle-free graph is a blow-up of its false-twin quotient, which is maximal triangle-free.*

Proof. A triangle in a blow-up projects to a triangle in the base, since classes are independent and adjacency between classes is complete or empty. A nonedge between classes $C_u \neq C_v$ comes from a base nonedge uv , whose common neighbour t lifts to the nonempty class C_t ; a nonedge inside a class C_u has a common neighbour in C_t for any base neighbour t of u , which exists since MTF graphs on ≥ 2 vertices have no isolated vertex. Conversely, vertices with equal open neighbourhoods are nonadjacent, classes are independent, and adjacency between classes is complete or empty, so G is a nonempty blow-up of the quotient H ; triangles in H lift to G , and common neighbours of nonadjacent representatives descend to H . $\square$

7 Main theorem

Theorem 7.1. *Assume Theorem 2.2. Then for every finite maximal triangle-free graph G the following are equivalent:*

1. $d_f(G) < 3$;
2. G satisfies D_4 ;
3. G is a blow-up of an Andrásfai or Vega graph.

Whenever these hold, $A(G)x = \mathbf{1}$ has a strictly positive rational solution. In particular Conjecture 1.1 is true.

Proof. (1) $\Rightarrow$ (2) is Lemma 3.1; (2) $\Leftrightarrow$ (3) is Theorem 2.2. For (3) $\Rightarrow$ (1) and the positive solution: by (4) and Theorem 5.1 every base graph has $d_f < 3$ and a positive rational equality solution; Lemma 6.1 transfers both along nonempty blow-ups (which are MTF by Lemma 6.2). $\square$

Corollary 7.2. *Under the same assumption, if $A(G)x = \mathbf{1}$, $x \geq 0$ is infeasible for a maximal triangle-free G , then $d_f(G) \geq 3$. Moreover $d_f(G) < 3$ forces a solution with no zero coordinate, and, within maximal triangle-free graphs, the first-order property D_4 coincides with the strict side of the LP threshold $d_f < 3$.*

Corollary 7.3. *Assume Theorem 2.2. Then Conjectures 4.1, 5.2, and 5.3 of [1] hold as well.*

Proof. For Conjecture 4.1: if $\delta(G) > n/3$ then $d_f(G) \leq n/\delta(G) < 3$, so Theorem 7.1 yields a positive rational solution of $Ax = \mathbf{1}$; clearing denominators gives positive integer multiplicities m with $Am = d\mathbf{1}$ for some integer d , and the corresponding vertex multiplication is a d -regular supergraph of G . Conjecture 5.2 ($d_f = \chi_f$ for maximal triangle-free graphs with $d_f < 3$) and Conjecture 5.3 (fractional-chromatic criticality of the cores in this class) are derived from Conjecture 5.1 in [1, Section 5], so they follow from Theorem 7.1. $\square$

8 A shorter route to Conjecture 1.1

The strong classification of [2] is not needed for Conjecture 1.1 alone: the ordinary Brandt–Thomassé structure theorem suffices. That theorem, proved in the unpublished manuscript [4] and recorded in [6] and [2, Section 1], states that every maximal triangle-free graph with minimum degree exceeding $n/3$ is a blow-up of an Andrásfai or Vega graph. (It is also a consequence of Theorem 2.2, since minimum degree exceeding $n/3$ implies D_k for every k [2].)

Theorem 8.1. *Assume the Brandt–Thomassé structure theorem. Then Conjecture 1.1 holds with a strictly positive solution, and Conjectures 4.1, 5.2, and 5.3 of [1] hold as well.*

Proof. Let G be maximal triangle-free with $d_f(G) < 3$, witnessed by a rational $p \geq 0$ with $Ap \geq \mathbf{1}$ and $\mathbf{1}^T p < 3$. Adding a small positive rational multiple of $\mathbf{1}$ preserves both constraints, so we may take $p > 0$. Choose a positive integer L with $m = Lp \in \mathbb{Z}_{>0}^{V(G)}$, and let H be the blow-up of G with class sizes m_v . Then $|V(H)| = \mathbf{1}^T m = L \mathbf{1}^T p < 3L$, while every vertex of H in the class of v has degree $(Am)_v = L(Ap)_v \geq L > |V(H)|/3$. By Lemma 6.2, H is maximal triangle-free, so the structure theorem makes H a blow-up of an Andrásfai or Vega graph. By (4), Theorem 5.1, and Lemma 6.1, H then carries a positive rational vector y with $A_H y = \mathbf{1}$. Finally, viewing H as a blow-up of G and summing y over its classes (Lemma 6.1 again) gives $x > 0$ with $A_G x = \mathbf{1}$. Conjecture 4.1 follows by clearing denominators as in Corollary 7.3, and Conjectures 5.2 and 5.3 by the derivations in [1, Section 5]. $\square$

This argument, which was pointed out during independent review of an earlier version of this note, shows that Conjecture 1.1 was in essence a consequence of previously available results, although we have not located a place where the deduction is stated. The role of the 2024 classification [2] in this note is therefore not to prove Conjecture 1.1, but to characterize *exactly* which maximal triangle-free graphs satisfy $d_f < 3$ (Theorem 7.1); the shorter argument gives no such converse.

9 Sharpness at $d_f = 3$

Let W be the graph with graph6 code H?q'qjo, i.e. $V = \{0, \dots, 8\}$ and

$$E(W) = \{04, 05, 08, 14, 17, 18, 25, 26, 28, 36, 37, 38, 46, 57\}.$$

Each neighbourhood is independent and every nonedge has a common neighbour, so W is maximal triangle-free, with degree sequence $(3^8, 4)$ and $\delta(W) = 3 = |V(W)|/3$.

The vector $p = (1, 0, 0, 1, 0, 0, 0, 1)^T$ satisfies $Ap \geq \mathbf{1}$, $\mathbf{1}^T p = 3$; the vector $z = \frac{1}{2}(1, 0, 0, 1, 1, 1, 1, 0)^T$ satisfies $A^T z = \mathbf{1}$, $\mathbf{1}^T z = 3$. Hence $d_f(W) = 3$ exactly. The complete nonnegative solution family of $Ax = \mathbf{1}$ is

$$x(t) = (t, \frac{1}{2} - t, \frac{1}{2} - t, t, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0)^T, \quad 0 \leq t \leq \frac{1}{2},$$

and the zero is forced: $r = \frac{1}{2}(0, 1, 1, 0, -1, -1, -1, -1, 2)^T$ satisfies $A^T r = e_8$ and $\mathbf{1}^T r = 0$, so every solution of $Ax = \mathbf{1}$ has $x_8 = r^T Ax = r^T \mathbf{1} = 0$.

Thus W has nonnegative equality solutions but no positive one: no vertex multiplication of W yields a regular graph. Since Conjecture 1.1 hypothesises the strict inequality $d_f(G) < 3$, and Brandt’s Conjecture 4.1 hypothesises minimum degree strictly exceeding $n/3$, the graph W (with $d_f = 3$ and $\delta = n/3$ exactly) contradicts nothing in [1]; rather, it shows both strict hypotheses are necessary, so the threshold in Theorem 7.1 is exact. The formulation recorded in [3] reads “minimum degree at least $n(G)/3$ ”; taken literally, that non-strict variant fails precisely at this boundary: W satisfies $\delta(W) = n(W)/3$ and admits no regular supergraph by vertex multiplications. The strict inequality in Brandt’s original statements is therefore essential and cannot be relaxed. All assertions about W are machine-checked by `verify/boundary9.py`.

10 Remarks

Computational corroboration. Independent exact-rational verification over all maximal triangle-free graphs of order at most 17 (catalogue counts matching the House of Graphs database) found no counterexample to Conjecture 1.1; every graph with $d_f < 3$ in that range admits a nonnegative equality solution, consistent with Theorem 7.1. The proof above does not depend on these computations.

What is and is not claimed. The result is conditional on [2]. The searches we performed located no prior statement of the equivalence in Theorem 7.1 or of the conjecture’s resolution, and [2] contains no mention of Conjecture 5.1, fractional domination, or LP duality; nevertheless we make no priority claim and request specialist review.

11 Provenance

The mathematics in this note was produced with substantial assistance from AI systems in research sessions directed by the author. Subsequently, every step was verified independently of those sessions: Lemma 3.1 by hand; the statements of Definition 2.1 and Theorem 2.2 against the posted text of [2]; Theorem 5.1 and Section 9 by machine, using the standalone scripts distributed with this note. A further independent review (also AI-assisted) identified the prior appearance of the Vega weights in [4, 5, 6] and supplied the shorter argument of Section 8; this version incorporates both corrections. The author takes full responsibility for the content.

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